



Seminar

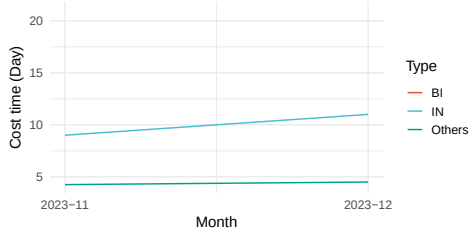
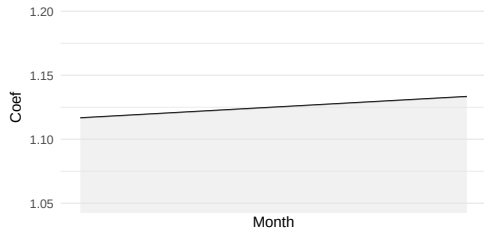
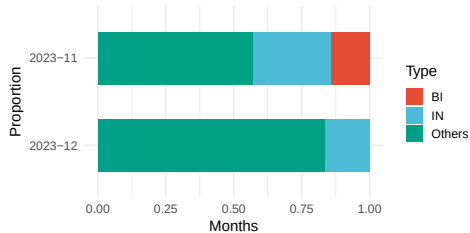
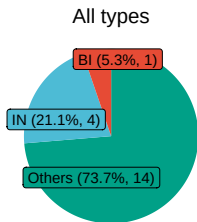
Lichuang Huang

Wie-Biotech

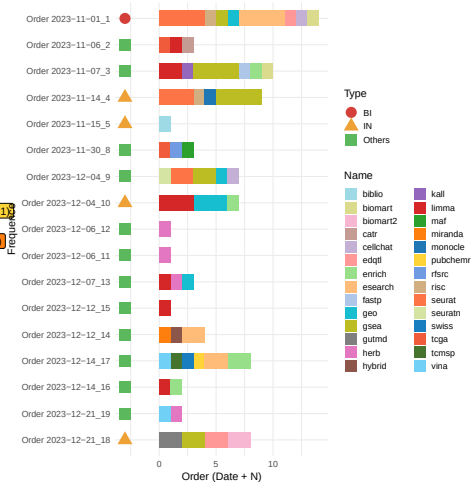
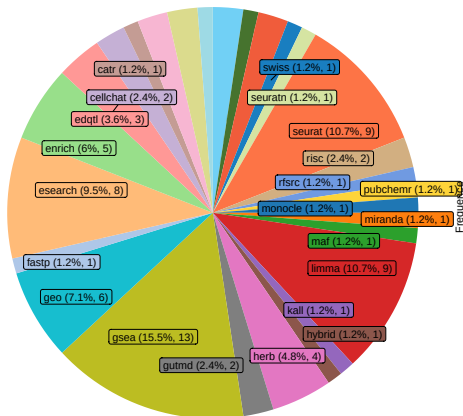
2024-01-12

Tasks summary

Overview



Methods used



New Methods

- edqtl: GTEX (eQTL, edQTL)
- biblio: bibliometrix
- hybrid: miRNA binding RNA
- miranda: miRNA binding mRNA
- tcmsp: TCM Database
- pubchemr: Compounds ID mapping
- swiss: Batch prediction
- biomart2: Biomart API mapping genes

Article sharing

RNA velocity

RNA velocity of single cells

Gioele La Manno^{1,2}, Ruslan Soldatov³, Amit Zeisel^{1,2}, Emelie Braun^{1,2}, Hannah Hochgerner^{1,2}, Viktor Petukhov^{3,4}, Katja Lidschreiber⁵, Maria E. Kastriti⁶, Peter Lönnerberg^{1,2}, Alessandro Furlan¹, Jean Fan³, Lars E. Borm^{1,2}, Zehua Liu³, David van Bruggen¹, Jimin Guo³, Xiaoling He⁷, Roger Barker⁷, Erik Sundström⁸, Gonçalo Castelo-Branco¹, Patrick Cramer^{5,9}, Igor Adameyko⁶, Sten Linnarsson^{1,2,†}, and Peter V. Kharchenko^{3,10,†}

¹Laboratory of Molecular Neurobiology, Department of Medical Biochemistry and Biophysics, Karolinska Institutet, 171 77 Stockholm, Sweden

²Science for Life Laboratory, 171 21 Solna, Sweden

³Department of Biomedical Informatics, Harvard Medical School, Boston, MA 02446, USA

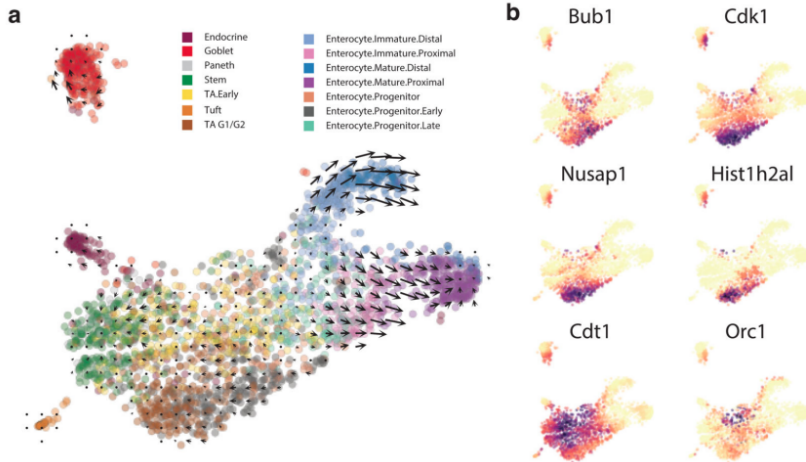
⁴Department of Applied Mathematics, Peter The Great St. Petersburg Polytechnic University, St. Petersburg, Russia

⁵Department of Biosciences and Nutrition, Karolinska Institutet, 171 77 Stockholm, Sweden

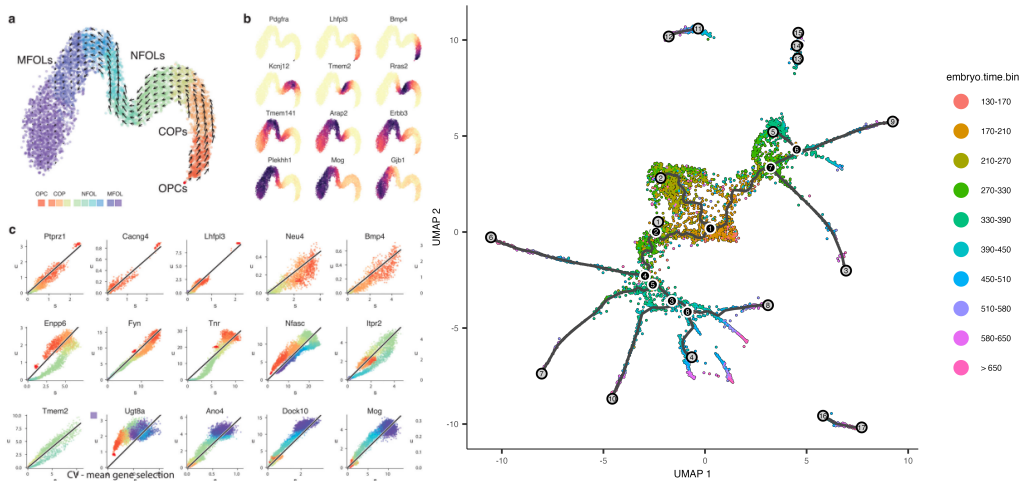
⁶Department of Physiology and Pharmacology, Karolinska Institutet, 171 77, Stockholm, Sweden

La Manno G, et al. (2018). *Nature*.

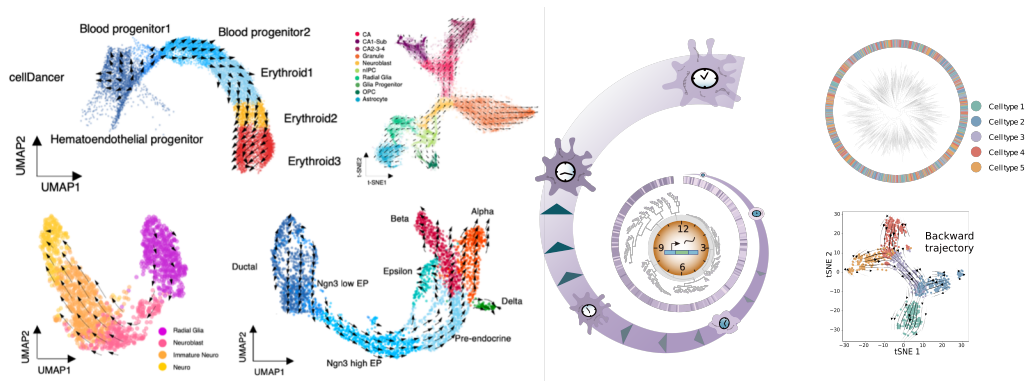
Details



Compare with Pseudotime analysis



Recent years: cellDancer and PhyloVelo



Schedule

Workflow document

2 Step 系列: scRNA-seq 基本分析

- 2.1 摘要
 - 2.1.1 目的
 - 2.1.2 解决的问题
- 2.2 方法
- 2.3 安装 (首次使用)
 - 2.3.1 安装依赖
 - 2.3.1.1 安装 Seurat v5
 - 2.3.1.2 安装 seurat-wrappers
 - 2.3.1.3 安装 monocle3
 - 2.3.1.4 安装 CellChat
 - 2.3.1.5 其它程序
 - 2.3.2 安装主体
- 2.4 使用说明
- 2.5 示例分析
 - 2.5.1 数据准备
 - 2.5.2 分析流程
 - 2.5.2.1 Job-seurat
 - 2.5.2.2 Step1
 - 2.5.2.3 Step2
- 2.6 技巧

2.3 安装 (首次使用)

2.3.1 安装依赖

2.3.1.1 安装 Seurat v5

Input

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
if (!requireNamespace("remotes", quietly = TRUE))
  install.packages("remotes")
install.packages(c("sva"))
BiocManager::install(c('SparseArray', 'fastDummies', 'RcppHNSW', 'RSpectra'))
remotes::install_github("satijalab/seurat", "seurat5")
```

2.3.1.2 安装 seurat-wrappers

Input

```
remotes::install_github('satijalab/seurat-wrappers')
```

2.3.1.3 安装 monocle3

Input

```
BiocManager::install(c('BiocGenerics', 'DelayedArray', 'DelayedMatrixStats',
```

Thanks